

Anshdeep Chaurasia

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EDUCATION

Indian Institute of Information Technology, Bhopal

Bachelor of Technology in Electronics and Communication Engineering

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CGPA - 8.43

PROJECTS

Hybrid Retrieval System for Financial QA | *LangChain, Pinecone, HuggingFace, BM25* | [GitHub](#)

- Implemented a hybrid search pipeline combining sparse (BM25) and dense (all-MiniLM-L6-v2) retrieval in a single Pinecone serverless index, with tunable alpha-weighted score blending.
- Added a cross-encoder reranking stage (bge-reranker-v2-m3) on top of hybrid retrieval, improving NDCG@10 from 66.3 (hybrid) to 70.5 (hybrid + rerank) — a 6% relative gain.
- Evaluated retrieval quality systematically against ground-truth relevance judgments from the BEIR FiQA-2018 financial QA benchmark, comparing sparse-only (NDCG@10: 39.1), dense-only (67.3), hybrid (66.3), and hybrid+rerank (70.5) configurations across 40 test queries.

DeepFake Video Detection System | *PyTorch, OpenCV, Streamlit, YOLO, MTCNN* | [GitHub](#)

- Developed an end-to-end DeepFake video detection system using EfficientNet-B4 for spatial feature extraction and a Transformer Encoder for temporal sequence modeling.
- Implemented and evaluated CNN-BiLSTM and CNN-Transformer architectures on the FaceForensics++ dataset; selected the Transformer model, achieving 88.67% test accuracy.
- Built a preprocessing pipeline using MTCNN for face detection and cropping, extracting 5 uniformly sampled frames per video to optimize computational efficiency.
- Created an interactive Streamlit dashboard with YOLO-based face tracking, confidence visualization, annotated video generation, and downloadable inference results.
- Evaluated model performance using accuracy, ROC curve, confusion matrix, classification report, and training history visualization.

Log Anomaly Detection & Monitoring System | *Python, Machine Learning, Data Processing, Streamlit* | [GitHub](#)

- Developed an anomaly detection system to identify unusual patterns in system/application logs.
- Implemented preprocessing pipeline for structured log parsing and anomaly detection.
- Applied machine learning techniques for anomaly detection and monitoring.
- Built a system capable of detecting suspicious or abnormal log activities.

Customer Segmentation using Machine Learning | *Python, Scikit-learn, Pandas, Streamlit* | [GitHub](#) | [Live Demo](#)

- Built a customer segmentation system using unsupervised machine learning techniques.
- Performed data preprocessing, feature analysis, and clustering for customer behavior analysis.
- Applied K-Means clustering to segment customers into behavioral groups.
- Visualized customer segments and extracted insights for targeted marketing strategies.

TECHNICAL SKILLS

Languages: Python, C/C++

ML/DL Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn

Fine-tuning & LLM Tools: LoRA, QLoRA, PEFT, Prompt Engineering

Retrieval & RAG: LangChain, Pinecone, Hybrid Search (BM25 + Dense Embeddings), Reranking

Computer Vision: OpenCV, YOLO, MTCNN, Vision Transformers (ViT)

Data & Visualization: NumPy, Pandas, Matplotlib, Streamlit

Developer Tools: Git, GitHub, Docker, Jupyter Notebook, PyCharm

RELEVANT COURSEWORK

- Data Structures & Algorithms
- Machine Learning
- Deep Learning
- Probability & Statistics